AI Blog REST API – Phase Document

Project Type: AI-powered Blog REST API

Phase 1 – Project Planning & Requirement Analysis

- Project title and objective
- Problem statement
- Existing system
- Proposed system
- Functional requirements
- Non-functional requirements
- Technology stack

Phase 2 – System Design

- System architecture
- API architecture
- Database design
- User roles: Admin, Editor, Author, Reader
- Authentication and authorization flow
- API endpoint design

Phase 3 – Backend Development

- Node.js and Express.js setup
- MongoDB connection
- Mongoose models
- User registration/login
- JWT authentication
- Password hashing using bcrypt
- Role-based access control

Phase 4 – AI Integration

- Gemini AI integration
- AI blog generation
- AI content improvement
- FAQ generation
- Prompt handling
- AI response processing

Phase 5 – REST API Development

- Authentication APIs
- Blog APIs
- AI APIs
- User APIs
- Category/tag APIs
- CRUD operations
- Error handling and validation

Phase 6 – Security & Optimization

- JWT security
- Rate limiting
- Input validation
- Password protection
- API error handling
- MongoDB query optimization

Phase 7 – Testing

- API testing using Postman
- Authentication testing
- Blog CRUD testing
- AI generation testing
- Error/validation testing
- Role-access testing

Phase 8 – Deployment & Documentation

- Environment configuration
- Database deployment
- Backend deployment
- API documentation
- Screenshots
- Final project report
- Future enhancements